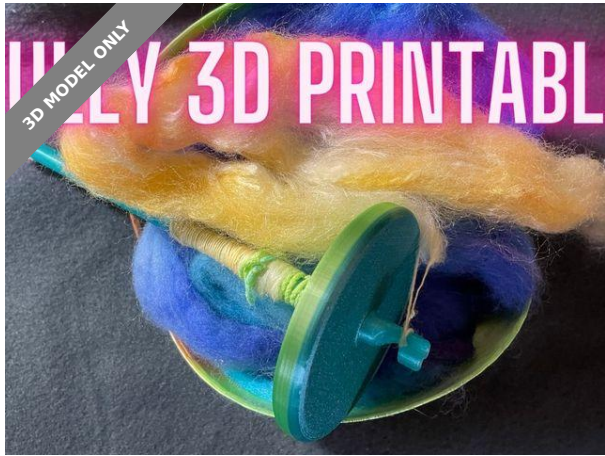




Fully printable drop spindle



Jazznow

[VIEW IN BROWSER](#)

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Summary

You want to make yarn from wool fibers? Here you go. Easy with this fully printable Drop Spindle.

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Tags: [spindle](#) [fullyprintable](#) [yarn](#) [wool](#) [yarncraft](#)
[handarbeit](#) [garn](#) [spindel](#) [dropspindle](#)

UPDATE:

- added instruction video in german (sorry no english. you dont want to hear my bad english for 20 minutes :D)
- added a whorl in single and dual color with tighter tolerances and a corresponding press fit test. Use those when you have a well calibrated printer
- added a stabilizer for the shaft. use it when you have problems with z-wobble. Import together with shaft so it will be aligned correctly. it will stabilize the upper part of the shaft, so the part that is important for the press fit will come out correctly

Ok, does your wife question the use of a 3d Printer at home? Does she love yarn, knitting, crocheting and stuff?

If the answer to both questions is yes: I've got some great news for you! I made the first (as far as I know)

Fully printable Drop Spindle

Just print her one (and if she doesn't spin her own yarn yet, buy her some wool fibers and show her the video I link below): She will never question the use of a 3d printer again. And if she does, she will be too busy with spinning and buying fibers to argue with you about it.

I call that a Win-Win.

My wife was so intrigued by that, she just bought her own 3D-Printer.

But be aware: Drop spindles are - just like 3D printers - herd animals. She will want a second and a third in no time :)

Features

- Easy and fully 3d printables
- Most of the time easy press fit (if your printer is dialed in).
- provided press fit test, so you can see if your printer is dialed in (I swear, my Neptune 3 is alive and just changes stuff at night, so prints I did before won't work the next day. Its to my fault...)
- Critical parts already provided with a pseudo-brim (because im forgetful as f*** and will forget to add a brim from time to time)
- Spindle works better than 60\$ wooden ones, my wife bought and costs only a few Cents (my wife said that, not I)
- All parts already oriented correctly
- no supports needed
- onecoloured and two colored version available

Printing instructions

For the whorl:

- use just standard 0.2mm profile
- print the files press-fit-test first. If it's too loose or too tight scale the outer ring a little bit and try again. If you found the right amount of scale, go to the next step.
 - if you have a printer, that's dialed in very good, use the version with tight tolerances, so you wont have to scale too much. make sure to use the correct whorl files too.
- print the whorl from PLA, scale it according to the result of the press-fit-test only in x- and y-direction, not in Z

- if you (or your wife) are an experienced spinner and want something super lightweight use LW-PLA. If you want something fancy (and who doesn't like fancy): rainbow silk looks soooo awesome with this.
- You can influence the weight of the spindle by playing around with the settings for walls, bottom and top layers and infill.
- if you print the 2 colored version import whorl-color1 and whorl-color2 together. Your slicer should orient them automatically (remember to scale both of them if you need to scale)
- if you want the the 1 colored version just print whorl

For the shaft and hook:

- use petg if you want it a little more stable. Both parts are a little delicate and can snap with only a slight amount of force if printed from PLA.
- be sure, that your Z-Wobble is under control! The shaft is thin and high, so it's prone to Z-wobble, but even if you get a bit of wobble, it still should work as intended.
- If your printer is prone to Z-Wobble you can use the provided stabilizer. It will stabilize the most upper part, so the section that is important for the press fit comes out correctly.

Assembly:

- just remove the brim from the shaft and hook.
- Clean up the edges of the hook with some fine sanding paper
- **THIS IS IMPORTANT:** The hook and the whorl need to be aligned! You'll find a little notch at the whorl. This notch needs to be aligned to the back of the hook (the closed part). If you don't do this, the yarn could slip.
- Take both parts, align them as described and press them into the whorl. If you scaled everything right, it should be tight (really tight).
- If it's too loose: just add some superglue.
- If it's too tight: Try sanding down the edges first and see if you can fit the part in. if that doesn't help, you may need to sand the whole thing or drill the hole in the whorl bigger. Or just do the press fit test again and print again
- Tada! You did it! Not that I had any doubt you could do it, but you did it! :D

Instructions Video for printing from Youtube (German)

Instruction video on how to spin from YouTube by Kristin Lehrer:

Video Anleitung Handspinnen auf deutsch von Chantimanou:

Model files



whorl-color2.stl

☐ for two colored whorl



whorl-color1.stl

☐ for two colored whorl



whorl.stl

☐ for single colored whorl



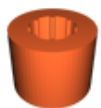
shaft.stl



hook.stl



press-fit-test-cylinder.stl



press-fit-test-ring.stl

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